A NORTHSTAR INC. TRAINING NARRATIVE

The People Analytics Story

Every question the CEO asked, every SQL query that answered it, and the twist nobody expected — from Week 2's first SELECT to a Week 4 recursive CTE.

Chapter 1

Chapter 2

Chapter 3

21 real SQL queries · tested against the live hr_training database

The Setup

You've just joined **Northstar Inc.** as a Data Analyst in People Analytics. It's July 2026. The CEO wants an **organizational health report** for the upcoming board meeting.

> ***"Start simple, then go deeper. Show me what the data says."***

— your manager, handing you database access

That's exactly what this course does — and by the end, you'll have answered the CEO's question, and found something nobody expected.

1. **Look at the data**
   Week 2 — raw SELECTs
2. **Compare & summarize**
   Week 3 — joins & KPIs
3. **Interrogate & act**
   Week 4 — subqueries, CTEs, DML

The Data You're Working With

Northstar's HR system spans 10 related tables — the story of a person's entire employment lifecycle.

`employees`
The hub — 500+ people, self-referencing manager_id

`departments`
10 business units, linked to locations

`jobs`
20 titles & salary bands

`salary_history`
Every raise & promotion

`performance_reviews`
1–5 star ratings, with a reviewer

`projects`
20 initiatives across departments

`employee_projects`
Who's staffed on what, and for how long

`leave_requests`
Type, dates, approval status

`attendance`
~20,000+ daily check-in/out rows

`locations`
5 cities Northstar operates in

employees is the hub almost every query runs through — and manager_id pointing back into itself is what makes recursive CTEs possible later.

CHAPTER 1

Just Show Me the Data

Week 2 — SQL Fundamentals

TOPICS COVERED

- SELECT statements
- ORDER BY & sorting
- Built-in functions
- WHERE & filtering
- Data types & conversion
- Logical operators

“Before we talk numbers, just show me who's in the system.”

What does an employee record actually look like?

Manager: “Before we talk numbers, just show me who's in the system.”

```
SELECT employee_id, first_name, last_name, hire_date, salary, employment_status
FROM employees
LIMIT 5;
```

RESULT & INSIGHT

employee_id 1 — Angel Hill — hired 2015, salary 320,000, the highest in the company. That's the CEO. Already, the raw data is telling a story before you've written a single filter.

Who's active in Engineering, best paid first?

Manager: “Okay — now show me who's actually active in Engineering, best paid first.”

```sql
SELECT first_name, last_name, salary
FROM employees
WHERE department_id = 1 AND employment_status = 'Active'
ORDER BY salary DESC
LIMIT 5;
```

RESULT & INSIGHT

Jeffrey Doyle tops Engineering at 239,729 — well clear of Craig Wise and Mason Jackson. WHERE narrows the rows, ORDER BY ranks what's left. Two clauses, and a raw table becomes a leaderboard.

Built-in functions — how old is everyone, really?

Northstar stores date_of_birth, not age — so age has to be calculated, not read.

```
SELECT first_name, last_name, date_of_birth,
       TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age
FROM employees
ORDER BY age DESC
LIMIT 5;
```

RESULT & INSIGHT

Keith Turner, 68, and Dawn Taylor, 67, top the list. TIMESTAMPDIFF() turns a stored date into a computed fact — the database doesn't store 'age' anywhere, SQL derives it on the fly, every time.

Logical operators — recent hires, or anyone on commission?

Two very different groups of people, one query, joined by OR.

```
SELECT first_name, last_name, hire_date, commission_pct
FROM employees
WHERE hire_date >= '2024-01-01' OR commission_pct > 0
LIMIT 5;
```

RESULT & INSIGHT

OR is inclusive, not exclusive — an employee only needs to satisfy one side. Swap it for AND and the result set could shrink dramatically, or even disappear. The choice of operator changes who shows up.

Data conversion — make it board-meeting friendly

Raw dates and exact-to-the-cent salaries don't belong in a CEO's slide deck.

```sql
SELECT first_name, last_name,
       DATE_FORMAT(hire_date, '%b %Y') AS hired,
       ROUND(salary/1000)*1000 AS salary_rounded
FROM employees
LIMIT 5;
```

RESULT & INSIGHT

Angel Hill, hired 'Aug 2015', salary rounded to 320,000. DATE_FORMAT and ROUND don't change the underlying data — they change how it's presented, which is exactly what a board slide needs.

NOT — who's not currently active?

The flip side of every WHERE clause you've written so far.

```sql
SELECT first_name, last_name, employment_status
FROM employees
WHERE employment_status <> 'Active'
LIMIT 5;
```

RESULT & INSIGHT

Anthony Robinson, Terminated. Barbara Bush, On Leave. <> (not equal) is the mirror image of every filter so far — sometimes the more useful question is who's excluded, not who's included.

Try It Yourself — 10 Questions

Fundamentals practice: SELECT, WHERE, ORDER BY, built-in functions, data conversion, and logical operators. Run these against hr_training.

1. List all employees hired after Jan 1, 2022, newest first.
2. Find all active employees in the Sales department.
3. Show employees earning between 50,000 and 80,000, highest first.
4. Calculate each employee's age today using date_of_birth.
5. Find employees hired in 2023 OR currently earning commission.
6. Format every hire_date as "Mon YYYY" (e.g. "Aug 2015").
7. List employees whose employment_status is NOT 'Active'.
8. Find employees whose first_name starts with the letter 'A'.
9. List all distinct job_id values used in the employees table.
10. Show employees whose last_name contains 'son'.

No answers on this slide, on purpose — open hr_training and try each one yourself before checking with a peer or instructor.

CHAPTER 1 TAKEAWAY

You can read the raw data.

6

SQL fundamentals covered, one manager question at a time

Six queries, six fundamentals — SELECT, WHERE, ORDER BY, built-in functions, logical operators, and data conversion. You can now pull any slice of Northstar's 500+ employees and shape how it looks.

But your manager didn't ask for a list. She asked for a report. Lists don't compare departments, and they don't reveal patterns. That's Chapter 2.